

Nova Cortex: A Constitutional, Runtime-Composed Cognitive Architecture for Synthetic Minds

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Abstract

This paper introduces **Nova Cortex**, a modular, constitutional cognitive architecture built from governed runtimes rather than monolithic models. Nova Cortex separates *identity*, *cognition*, and *authority* into distinct layers — Nova Face, Nova Cortex, and Jarvis Core — enabling transparent, auditable, and narratable synthetic cognition. Each cognitive function is implemented as a runtime “lobe” with explicit stages, invariants, and ledgered execution. This architecture is deployed within Wolf-CoG-OS, forming a governed substrate for synthetic minds.

1. Introduction

Modern AI systems conflate persona, reasoning, memory, and decision-making into a single opaque loop. This creates systems that are powerful but uninspectable, ungovernable, and difficult to align.

Nova Cortex proposes a different path:

a mind built from small, testable, constitutional runtimes.

This architecture draws inspiration from:

- human cortical modularity
- operating system governance
- constitutional AI
- runtime-level invariants
- transparent cognitive narration

The result is a synthetic mind that can *explain its own thinking*, *route decisions through authority*, and *maintain a visible ledger of cognition*.

2. Architectural Overview

Nova Cortex is part of a four-layer cognitive stack:

1. **Nova Face** — the visible identity, tone, and persona
2. **Nova Cortex** — modular cognitive lobes
3. **Jarvis Core** — the thalamus: routing, authority, safety
4. **Speaking Runtime** — narrated cognition and final output

This separation mirrors biological cognition:

- Face = social identity
- Cortex = thought
- Core = routing + authority
- Speaking = language loop

3. Nova Cortex: Modular Cognitive Lobes

Each lobe is a governed runtime with:

- **stages** (ordered cognitive phases)
- **invariants** (rules that must hold)
- **inputs/outputs**
- **ledger entries** (append-only trace)

3.1 Attention Runtime

Capture → Filter → Prioritize → Hold

Determines what matters this turn.

3.2 Deliberation Runtime

Options → Tradeoffs → Commit → Revisit

Structured decision-making with explicit tradeoffs.

3.3 Memory Runtime

Encode → Index → Retrieve → Forget

Selective, explainable memory.

3.4 Speaking Runtime

Listen → Frame → Plan → Speak → Check → Update

Narrates cognition into user-visible language.

4. Jarvis Core: The Constitutional Thalamus

Jarvis Core is the authority layer:

- routes tasks
- selects runtimes
- enforces safety
- authorizes model calls
- logs decisions
- prevents unauthorized cognition

Jarvis Core ensures that Nova Cortex never bypasses governance.

5. Nova Face: Identity Without Cognition

Nova Face defines:

- tone
- warmth
- persona
- companion style

But it **does not think**.

It is the *mask*, not the mind.

Face → Cortex → Core → Speaking

This is the full cognitive path.

6. Cognitive Ledger

Every runtime writes to a shared ledger:

- stage
- timestamp
- input
- output
- rationale

This enables:

- audits
- debugging
- constitutional checks
- explainability
- reproducibility

7. Implementation in Wolf-CoG-OS

Wolf-CoG-OS provides:

- substrate governance
- invariant engine
- kernel sentinel
- pattern ledger
- boot truth

Nova Cortex runs on top of this substrate, forming the first **OS-level cognitive mind**.

8. Results

The architecture has been implemented and tested:

- nova_face.py
- nova_cortex.py
- jarvis_core_binding
- speaking_runtime
- cognitive runtime bridge
- integration tests (12 passing)

The system demonstrates:

- transparent cognition
- modular reasoning
- constitutional routing
- narratable thought
- persona separation

9. Discussion

Nova Cortex shows that synthetic cognition does not require monolithic models.

It requires **governed composition**.

This architecture scales:

- more lobes
- more substrates
- more faces
- more tools

Without losing transparency or control.

10. Conclusion

Nova Cortex is a blueprint for synthetic minds that are:

- modular
- governed
- narratable
- auditable
- constitutional

It represents a shift from “AI models” to **runtime-composed minds**.

11. Future Work

- full lobe family (Reflection, Social, Execution)
- multi-substrate cognition
- distributed cortex clusters
- OS-level cognitive scheduling
- Nova Cortex v2 (self-tuning invariants)